

Ideation Phase

Brainstorm & Idea Prioritization

Date	18 June 2026
Student Name	INAPAKUTHIKA BHARATH VARMA (bharathvarma906@gmail.com)
Team ID	
Project Name	ShopEz
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Using a structured brainstorming approach allows for the efficient discovery of software functional needs and high-level system architecture. This stage emphasizes ranking key functionalities for the ShopEz MERN ecosystem while verifying that technical requirements align with the project's practical boundaries.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Selection of the Problem Statement

Problem Identification: Current digital shoppers seek a seamless, responsive, and accelerated purchasing journey across varying platforms. Many existing e-commerce systems lack a cohesive administrative interface or the ability to facilitate real-time logistics tracking within a unified technology stack.

Selected Problem Statement: "In what way can we engineer a MERN-based full-stack solution that ensures an effortless product discovery and payment journey for customers, while simultaneously providing a robust and protected dashboard for administrative oversight?"

Step-2: Brainstorm, Idea Listing and Grouping

Classification of primary concepts derived from the individual ideation workshop:

User Interface (Frontend)

- React-driven product display grid
- Adaptive styling for diverse categories
- Live search and granular filter tools
- Global state management for cart (Redux)

Backend & Logic

- Token-based (JWT) secure sign-in
- Modular Express.js API design
- Package tracking and audit logging
- Automated stock level adjustments

Database (MongoDB)

- Optimized schemas for core entities
- Performance-tuned category indexing
- Cloud-based media asset handling

Admin Features

- Restricted Administrative Console
- Inventory management (CRUD)
- Order lifecycle processing

Step-3: Idea Prioritization

Assessment of project modules against Utility and Practicality scales to guarantee an effective Minimum Viable Product launch.

Feature Category	Importance (High/Med)	Feasibility (High/Med)	Priority Level
MERN Core & Database Configuration	High	High	Critical (Phase 1)
Secure Auth (User & Admin)	High	High	Critical (Phase 1)
Admin Management Tools	High	Medium	High (Phase 2)
Search & Discovery Algorithms	Medium	High	Standard (Phase 2)
Live Order Status Updates	Medium	Medium	Secondary (Phase 3)